

React JS Hands-on Lab

ES6 Features – map(), Arrow Functions & Destructuring (Cricket App)

Objective

Create a React Application named **cricketapp** with components that use ES6 features such as the **map()** method, arrow functions and destructuring.

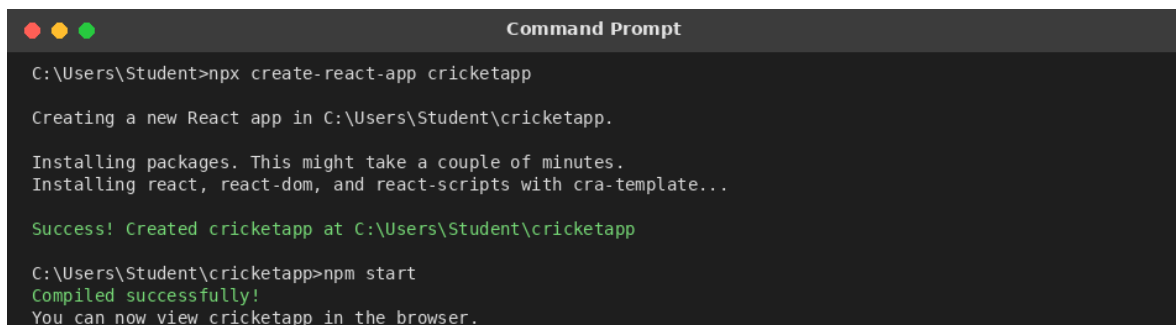
In this lab I have learnt how to:

- Use map() method of ES6
- Apply arrow functions of ES6
- Implement Destructuring features of ES6

Step 1: Create the React Project

Created a new React project named **cricketapp** and started the development server:

```
npx create-react-app cricketapp
cd cricketapp
npm start
```



```
Command Prompt

C:\Users\Student>npx create-react-app cricketapp

Creating a new React app in C:\Users\Student\cricketapp.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

Success! Created cricketapp at C:\Users\Student\cricketapp

C:\Users\Student\cricketapp>npm start
Compiled successfully!
You can now view cricketapp in the browser.
```

Screenshot: create-react-app and npm start output

Step 2: ListofPlayers Component (map)

Declared an array of 11 players with their names and scores, and used the ES6 **map()** method to display the list:

```
// src/components/ListofPlayers.js
export function ListofPlayers(props) {
  const players = props.players;
  return (
    <div>
      {
        players.map((item) => {
          return (
            <div>
              <li>Mr. {item.name} <span> {item.score} </span></li>
            </div>
          )
        })
      }
    </div>
  )
}
```

```
}
```

Code: src/components/ListofPlayers.js

Step 3: Scorebelow70 Component (Arrow Functions)

Used an ES6 arrow function inside **map()** to filter the players whose score is below 70:

```
// src/components/Scorebelow70.js
export function Scorebelow70(props) {
  const players = props.players;
  let players70 = [];
  players.map((item) => {
    if (item.score <= 70) {
      players70.push(item);
    }
  })
  return (
    <div>
      {
        players70.map((item) => {
          return (
            <div>
              <li>Mr. {item.name} <span> {item.score} </span></li>
            </div>
          )
        })
      }
    </div>
  )
}
```

Code: src/components/Scorebelow70.js

Step 4: IndianPlayers Component (Destructuring & Merge)

Used ES6 array destructuring (skipping elements) to display the Odd Team and Even Team players, and used the ES6 spread/merge feature to combine the T20Players and RanjiTrophyPlayers arrays:

```
// src/components/IndianPlayers.js
export function OddPlayers([first, , third, , fifth]) {
  return (
    <div>
      <li> First: {first} </li>
      <li> Third : {third} </li>
      <li> Fifth : {fifth}</li>
    </div>
  )
}

export function EvenPlayers([, second, , fourth, , sixth]) {
  return (
    <div>
      <li> Second : {second} </li>
      <li> Fourth : {fourth} </li>
      <li> Sixth : {sixth}</li>
    </div>
  )
}

const T20Players = ['First Player', 'Second Player', 'Third Player'];
const RanjiTrophyPlayers = ['Fourth Player', 'Fifth Player', 'Sixth Player'];
```

```

export const IndianPlayers = [...T20Players, ...RanjiTrophyPlayers];

export function ListofIndianPlayers(props) {
  return (
    <div>
      {
        props.IndianPlayers.map((item) => {
          return (
            <li>Mr. {item}</li>
          )
        })
      }
    </div>
  )
}

```

Code: src/components/IndianPlayers.js

Step 5: App.js – Rendering with a Flag Variable

Declared the players data and used a simple if-else on the **flag** variable in **App.js** to display either the ListofPlayers/Scorebelow70 components or the IndianPlayers components on the same home page:

```

// src/App.js
import './App.css';
import { ListofPlayers } from './components/ListofPlayers';
import { Scorebelow70 } from './components/Scorebelow70';
import { OddPlayers, EvenPlayers, IndianPlayers, ListofIndianPlayers } from './components/IndianPlayers';

const players = [
  { name: 'Jack', score: 50 },
  { name: 'Michael', score: 70 },
  { name: 'John', score: 40 },
  { name: 'Ann', score: 61 },
  { name: 'Elisabeth', score: 61 },
  { name: 'Sachin', score: 95 },
  { name: 'Dhoni', score: 100 },
  { name: 'Virat', score: 84 },
  { name: 'Jadeja', score: 64 },
  { name: 'Raina', score: 75 },
  { name: 'Rohit', score: 80 },
];

const IndianTeam = ['Sachin1', 'Dhoni2', 'Virat3', 'Rohit4', 'Yuvraj5', 'Raina6'];

function App() {
  var flag = true;
  if (flag === true) {
    return (
      <div>
        <h1> List of Players</h1>
        <ListofPlayers players={players} />
        <hr />
        <h1> List of Players having Scores Less than 70 </h1>
        <Scorebelow70 players={players} />
      </div>
    )
  }
  else {
    return (
      <div>

```

```

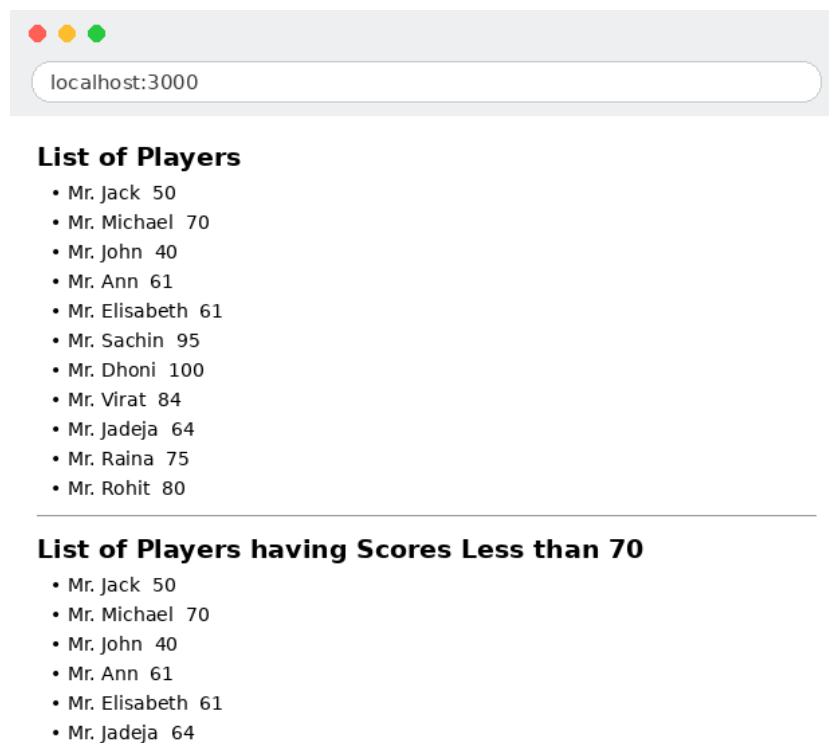
    <div>
      <h1> Indian Team </h1>
      <h1> Odd Players </h1>
      {OddPlayers(IndianTeam)}
      <hr />
      <h1> Even Players</h1>
      {EvenPlayers(IndianTeam)}
    </div>
  <hr />
  <div>
    <h1> List of Indian Players Merged:</h1>
    <ListofIndianPlayers IndianPlayers={IndianPlayers} />
  </div>
</div>
)
}
}

export default App;

```

Code: src/App.js

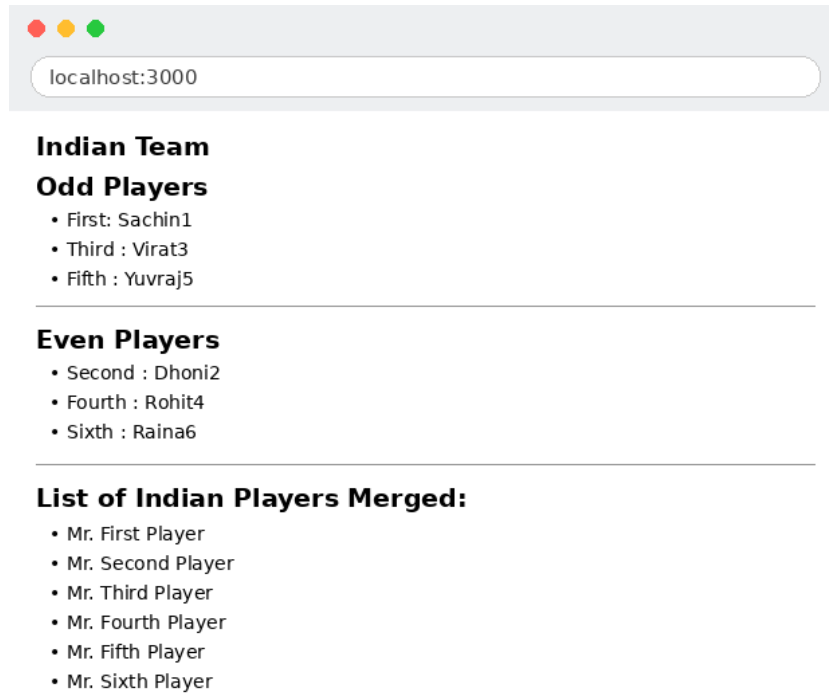
Output: When Flag = true



Screenshot: Output rendered at localhost:3000 when flag = true

Output: When Flag = false

Set **var flag = false;** in App.js to view the second output:



Screenshot: Output rendered at localhost:3000 when flag = false

End of Lab